

Schema Design

- **country** (
 country_name
)
- **language** (
 language_name
)
- **genre** (
 genre_name
)
- **mood** (
 mood_name
)
- **instrument** (
 instrument_name
)
- **gender** (
 gender_type
)
- **target** (
 target_id,
 target_type
)
- **user** (
 user_id,
 target_id FK references target (target_id), (1,1) to (0,1)
 user_type {admin, listener, artist},
 first_name,
 last_name,
 email,
 password,
 profile_picture,
 bio,
 country FK references country (country_name), (1,1) to (0, N)
 language FK references language (language_name), (1,1) to (0, N)
 phone number,
 gender FK references (gender_type), (1,1) to (0, N)
 date_of_birth,
 app_mode {light, dark}
)

- **listener** (
listener_id FK references user (user_id), (1,1) to (0,1)
listener_type {free listener, premium listener},
)
- **artist** (
artist_id FK references user (user_id), (1,1) to (0,1)
stage_name,
bank_account,
points
)
- **admin** (
admin_id FK references user (user_id), (1,1) to (0,1)
role {super admin, moderator, analyst, audit}
)
- **language_preference** (
listener_id FK references listener (listener_id), (1,1) to (0, N)
language FK references language (language_name) (1,1) to (0, N)
)
- **followed_artist** (
listener_id FK references listener (listener_id), (1,1) to (0, N)
artist_id FK references artist (artist_id) (1,1) to (0, N)
)
- **friend** (
user1_id FK references listener (listener_id) (1,1) to (0, N)
user2_id FK references listener (listener_id) (1,1) to (0, N)
)
- **friend_request** (
sender_id FK references listener (listener_id), (1,1) to (0, N)
receiver_id FK references listener (listener_id), (1,1) to (0, N)
status{accepted, pending, rejected},
date_time
)
- **song** (
song_id,
target_id FK references target (target_id) (1,1) to (0,1)
title,
album_id FK references album (album_id), (1,1) to (1, N)
owner_id FK references artist (artist_id), (1,1) to (0,N)
language FK references language (language_name), (1,1) to (0, N)
length,
release_date,
song_audio,

- lyrics,
 - copyright certificate
-)
- **album** (
 - album_id,
 - target_id FK references target (target_id), (1,1) to (0,1)
 - title,
 - description,
 - release date,
 - cover_picture,
 - copyright_certificate
-)
- **playlist** (
 - playlist_id,
 - target_id FK references target (target_id) (1,1) to (0,1)
 - title,
 - creator_id FK references listener (listener_id), (1,1) to (0, N)
 - description,
 - creation_date,
 - visibility {public, private},
 - cover_picture
-)
- **song_artist** (
 - song_id FK references song (song_id), (1,1) to (1, N)
 - artist_id FK references artist (artist_id), (1,1) to (0, N)
 - role {singer, lyricist, composer}
-)
- **song_genre** (
 - song_id FK references song (song_id), (1,1) to (1, N)
 - genre FK references genre(genre_name)(1,1) to (0,N)
-)
- **song_mood** (
 - song_id FK references song (song_id), (1,1) to (1, N)
 - mood FK references mood (mood_name) (1,1) to (0, N)
-)
- **song_instrument** (
 - song_id FK references song (song_id), (1,1) to (1, N)
 - instrument FK references instrument (instrument_name) (1,1) to (0, N)
-)
- **playlist_song** (
 - playlist_id FK references playlist (playlist_id), (1,1) to (0, N)
 - song_id FK references song (song_id) (1,1) to (0, N)

-)
- **song_stream_history** (
 - stream_id,
 - listener_id FK references listener (listener_id), (1,1) to (0,N)
 - song_id FK references song (song_id), (1,1) to (0,N)
 - date_time,
 - duration
-)
- **album_stream_history** (
 - stream_id,
 - listener_id FK references listener (listener_id), (1,1) to (0,N)
 - album_id FK references album (album_id), (1,1) to (0,N)
 - date_time,
 - duration
-)
- **playlist_stream_history** (
 - stream_id,
 - listener_id FK references listener (listener_id), (1,1) to (0,N)
 - playlist_id FK references playlist (playlist_id), (1,1) to (0,N)
 - date_time,
 - duration
-)
- **liked_song** (
 - listener_id FK references listener (listener_id), (1,1) to (0,N)
 - song_id FK references song (song_id) (1,1) to (0,N)
-)
- **liked_album** (
 - listener_id FK references listener (listener_id), (1,1) to (0,N)
 - album_id FK references album (album_id) (1,1) to (0,N)
-)
- **liked_playlist** (
 - listener_id FK references listener (listener_id), (1,1) to (0,N)
 - playlist_id FK references playlist (playlist_id) (1,1) to (0,N)
-)
- **transaction_history** (
 - transaction_id,
 - user_id FK references user (user_id), (1,1) to (0,N)
 - transaction_type {subscription, buy, withdraw, refund},
 - amount,
 - payment method {bank, bkash,...},
 - date_time,
 - status

-)
- **plan**(
 - plan_id,
 - plan_type,
 - plan_cost,
 - max_members
-)
- **plan_subscription** (
 - subscription_id,
 - plan_id FK references plan (plan_id), (1,1) to (0,N)
 - owner_id FK references listener (listener_id), (1,1) to (0,N)
 - start_date,
 - end_date,
 - transaction_id FK references transaction (transaction_id) (1,1) to (0,1),
 - auto-renewal_mode {on, off}
-)
- **family_plan** (
 - family_plan_id,
 - parent_account_id FK references listener (listener_id), (1,1) to (0, N)
 - subscription_id FK references subscription (plan_subscription_id) (1,1) to (0, N)
-)
- **family_plan_member** (
 - family_plan_id FK references family_plan (family_plan_id), (1,1) to (1,N(5/6))
 - member_id FK references listener (listener_id) (1,1) to (0,N)
-)
- **friend_shared_content** (
 - shared_id,
 - sender_id FK references listener (listener_id), (1,1) to (0,N)
 - receiver_id FK references user (user_id), (1,1) to (0, N)
 - content_type {song, playlist},
 - content_id FK references target (target_id), (1,1) to (0,N)
 - date_time
-)
- **family_shared_content** (
 - shared_id,
 - family_plan_id FK references family_plan (family_plan_id), (1,1) to (0,N)
 - content_type {song, playlist},
 - content_id FK references target (target_id), (1,1) to (0,N)
 - date_time
-)
- **product** (
 - product_id,

target_id FK references target (target_id), (1,1) to (0,1)
category {ticket, merch, CD,...},
product_name,
owner_id FK references artist (artist_id), (1,1) to (0,N)
description,
product_image,
price,
stock_quantity,
lifetime

-)
- **cart** (
 - cart_id,
 - owner_id FK references listener (listener_id), (1,1) to (0,N)
 - transaction_id FK references transaction (transaction_id), (1,1)to (0,1)
 - date_time)
- **cart_items** (
 - cart_id FK references cart (cart_id), (1,1) to (0,N)
 - product_id FK references product (product_id), (1,1) to (0,N)
 - quantity)
- **badge** (
 - badge id,
 - badge_name,
 - description,
 - image,
 - artist_id FK references artist (artist_id), (1,1) to (0,N))
- **badge_user** (
 - user_id FK references user (user_id), (1,1) to (0,N)
 - badge_id FK references badge (badge_id), (1,1) to (0,N)
 - date_time)
- **approval_request** (
 - request id,
 - request_type {artist, song},
 - content_id FK references target (target_id) (1,1) to (0,1))
- **announcement** (
 - announcement id,
 - announcer_id FK references user (user_id), (1,1) to (0,N)
 - text,
 - image,)

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        date_time
    )
    • review (
        review_id,
        reviewer_id FK references listener (listener_id), (1,1) to (0,N)
        topic {system, song, playlist, artist},
        topic_id FK references target (target_id), (1,1) to (0,N)
        text,
        image,
        date_time
    )
    • report (
        report_id,
        author_id FK references user (user_id), (1,1) to (0,N)
        topic {system, song, playlist, album, listener, artist, product}
        target_id FK references target (target_id), (1,1) to (0,N)
        text,
        image,
        date_time
    )

```